

CRISPR gRNA Design Report — BlaCTX AMR Gene

Objective

To design and validate guide RNAs targeting the BlaCTX antimicrobial resistance gene using Cas9, Cas12 and Cas13 through computational tools.

What I Did

Step 1 — Sequence Retrieval

I downloaded the BlaCTX gene sequence from NCBI GenBank. The accession number is CP162271.1 from *Kluyvera sichuanensis*. The sequence length is 876 bp.

Step 2 — Sequence Visualization

I imported the FASTA sequence into SnapGene Viewer to visualize the gene and identify suitable regions for gRNA targeting.

Step 3 — gRNA Design

I used CHOPCHOP v3 to design 3 gRNAs for each Cas type:

- Cas9 — requires NGG PAM, gRNA length is 20nt
- Cas12 — requires TTTV PAM, gRNA length is 28nt
- Cas13 — no strict PAM, uses PFS, gRNA length is 28nt, targets RNA

Step 4 — Annotation

All 9 gRNA sequences along with their PAM sites were manually annotated on the BlaCTX sequence in SnapGene. Each Cas type was color coded separately for clarity.

Step 5 — Validation

Top gRNA from each Cas type was validated using NCBI BLAST against the nucleotide collection database. All three showed 100% identity match to the source organism with zero off-target hits.

Observations

- All 9 gRNAs showed zero off-targets (MM0) which means high specificity
- Cas9 gRNA1 had the highest efficiency score of 63.46
- Cas12 gRNA3 showed low GC content of 29% which is outside the ideal range of 40-70%, so Cas12 gRNA1 is the better candidate

- Cas13 gRNAs all clustered around position 101-104 indicating a highly accessible region on the RNA transcript

Conclusion

All three Cas systems successfully produced specific gRNAs targeting the BlaCTX gene. Cas9 gRNA1 is the best candidate for gene editing applications.